

(Task 1 of 4) In this tutorial, we will review previously learned content.

0

(Task 2 of 4) What is the result of running this program?

Pseudo | Pseudo [Run 

JavaScript

```

fun k(a):          function k(a) {
  let b = 1        let b = 1;
  return a + b     return a + b;
end               }
print(k(3) + b)   console.log(k(3) + b);

```

error

2

You predicted the output correctly 🎉🎉🎉

3

`k(3) + b` is evaluated in the top-level block, where `b` is not defined. So, this program errors.

Click [here](#) to run this program in the Stacker.

(Task 3 of 4) What is the result of running this program?

Pseudo | Pseudo [Run 

Scala 3

```

let i = j * 3    val i = j * 3
let j = 2        val j = 2
print(i)         println(i)
print(j)         println(j)

```

error

5

You predicted the output correctly 🎉🎉🎉

6

The first definition tries to bind `i` to the value of `j * 3`. To evaluate `j * 3`, we need the value of `j`. But `j` is not bound to a value at that moment.

Click [here](#) to run this program in the Stacker.

(Task 4 of 4) What is the result of running this program?

Pseudo | Pseudo [Run 

Lispy

```

let a = 3          (defvar a 3)
fun foo(b):        (deffun (foo b)
  fun bar():       (deffun (bar)

```

```
run bar():          (defun (bar)
  let c = 6          (defvar c 6)
  return a + b + c    (+ a b c))
end                  (bar))
return bar()          (+ (foo 4) 2)
end
print(foo(4) + 2)
```

15

8

You predicted the output correctly 🎉🎉🎉

9

This program binds `a` to 3 and `foo` to a function, and then evaluates `foo(4) + 2`. The value of `foo(4) + 2` is the value of `bar()` + 2, which is the value of `(a + b + z) + 2`, which is the value of `(3 + 4 + 6) + 2`, which is 15.

Click [here](#) to run this program in the Stacker.

You have finished this tutorial 🎉🎉🎉

Please the finished tutorial to a PDF file so you can review the content in the future. **Your instructor (if any) might require you to submit the PDF.**

Start time: 1789959889841